

Reviewer Pack — Minimal Rank of Noncrossing Partition Matrices vs Monotone C...

Entry a5627e06dd88 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Noncrossing Partition Matrices vs Monotone Circuit Depth for k-CLIQUE

Entry ID: a5627e06dd88 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-24 21:03:00 UTC

1. Conjecture

Field A (mathematical branch): Combinatorial Geometry (Noncrossing Partitions) **Field B** (complexity object): Complexity Theory: Monotone Circuit Complexity (for k-CLIQUE)

Statement:

[‘For any monotone circuit C of size n that computes the OR of all edges in a complete graph on k vertices, the minimal rank of the noncrossing partition matrix associated with C is $\Omega(n^{(1/4+k/16)})$.’, ‘Additionally, for any DNF formula representing the OR of all edges in such a complete graph, the minimal rank of its noncrossing partition matrix is $O(n^2 \log n)$.’]

Rationale (proposer’s reasoning):

[‘Noncrossing partitions are a combinatorial structure with strong geometric interpretations and have been used in various areas of mathematics. They provide a way to encode boolean functions into matrices, which can potentially be exploited for proving lower bounds on monotone circuit complexity.’, “The conjecture builds upon the idea that the noncrossing partition matrix captures essential information about the circuit’s ability to compute k-CLIQUE, leveraging geometric properties not typically utilized in complexity theory.”]

Taxonomy category: MONOTONE_CLIQUE (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 2501daaf3483f3b2

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The minimal rank of noncrossing partition matrices associated with monotone circuits is compared to the size of the circuits.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	SAFE
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	1.00	SAFE	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 4 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - noncrossing partitions monotone circuit depth k-clique - minimal rank noncrossing partition matrices complexity theory OR edges - combinatorial geometry non-crossing partitions monotone circuits k-CLIQUE

Top relevant hits considered: - [http://arxiv.org/abs/1604.06009v2] Oriented Flip Graphs and Noncrossing Tree Partitions - [http://arxiv.org/abs/2212.13799v6] Noncrossing partitions of a marked surface - [http://arxiv.org/abs/1904.05573v3] k -Indivisible Noncrossing Partitions - [http://arxiv.org/abs/1509.06942v2] Symmetric Decompositions and the Strong Sperner Property for Noncrossing Partition Lattices

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def noncrossing_partition_matrix(circuit):
        n = len(circuit)
        matrix = [[0] * n for _ in range(n)]
        for i in range(n):
            u, v = map(int, circuit[i][4:].split(", "))
            if 0 <= u < n and 0 <= v < n:
                matrix[u][v] = 1
                matrix[v][u] = 1
        return matrix

    def rank(matrix):
        m, n = len(matrix), len(matrix[0])
        for i in range(m):
            if matrix[i][i] == 0:
                found = False
                for j in range(i + 1, m):
                    if matrix[j][i] != 0:
                        matrix[i], matrix[j] = matrix[j], matrix[i]
```

```

        found = True
        break
    if not found:
        return i
    for j in range(n):
        if j != i and matrix[i][j] != 0:
            factor = matrix[j][i] / matrix[i][i]
            for k in range(n):
                matrix[j][k] -= factor * matrix[i][k]
    return m

def generate_k_clique(k, n):
    edges = []
    for i in range(n):
        for j in range(i + 1, n):
            if len(edges) >= k * (k - 1) // 2:
                break
            edges.append((i, j))
        if len(edges) >= k * (k - 1) // 2:
            break
    return edges

def monotone_circuit(k):
    n = k * (k - 1) // 2
    circuit = []
    for i in range(n):
        u, v = random.sample(range(k), 2)
        if u > v:
            u, v = v, u
        circuit.append(f"OR({u},{v})")
    return circuit

k_values = [5, 10, 15, 20, 30, 40]
results = []

for k in k_values:
    n = k * (k - 1) // 2
    clique_edges = generate_k_clique(k, n)
    circuit = monotone_circuit(k)

    if len(circuit) != n:
        return {
            "metric_name": "rank",
            "metric_value": None,
            "instances_tested": 0,
            "conjecture_holds": False,
            "counterexample": "circuit_size_mismatch"
        }

    matrix = noncrossing_partition_matrix(circuit)
    rank_value = rank(matrix)

    results.append({
        "k": k,
        "n": n,
        "rank": rank_value
    })

```

```

    })

    if not results:
        return {
            "metric_name": "rank",
            "metric_value": None,
            "instances_tested": 0,
            "conjecture_holds": False,
            "counterexample": "no_results"
        }

    mean_rank = sum(result["rank"] for result in results) / len(results)
    std_rank = math.sqrt(sum((result["rank"] - mean_rank) ** 2 for result in results) / len(results))

    return {
        "metric_name": "rank",
        "metric_value": mean_rank,
        "instances_tested": len(results),
        "conjecture_holds": True,
        "counterexample": ""
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3

    results = []

    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    if all(result["conjecture_holds"] for result in results):
        mean_rank = sum(result["metric_value"] for result in results) / len(results)
        std_rank = math.sqrt(sum((result["metric_value"] - mean_rank) ** 2 for result in results) / len(results))
        print(f"RESULT: SUPPORTED mean={mean_rank} std={std_rank} support_fraction=1.0")
    else:
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample='rank_mismatch' first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a3aa7e17.py", line 124, in <module>
    trial_result = run_trial(seed)
    ~~~~~^~~~~~
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a3aa7e17.py", line 88, in run_trial
    matrix = noncrossing_partition_matrix(circuit)
    ~~~~~^~~~~~

```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a3aa7e17.py", line 25, in noncrossing_p
    u, v = map(int, circuit[i][4:].split(", "))
    ^^^^
```

ValueError: invalid literal for int() with base 10: ',4)'

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed before producing data, which means that it was unable to complete the computation and verify the conjecture. | next: Investigate the cause of the crash in the test code and attempt to run the test again. If the crash is resolved, re-evaluate the conjecture based on the results.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	11618
2	propose	ollama_remote	glm4:latest	0	0	10930
3	preregistration	ollama_remote	glm4:latest	0	0	5300
4	novelty	ollama_remote	glm4:latest	0	0	4593
5	novelty	ollama_remote	glm4:latest	0	0	6243
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	16474
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11377
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12721
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12188
10	judge	ollama_remote	glm4:latest	0	0	8421

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 99866 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in *research/audit/a5627e06dd88.jsonl*)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in *research/audit/a5627e06dd88.jsonl* for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: *research/pvsnp_sandbox/test_*.py* (per-cycle)

Replay tarball: *research/replay/a5627e06dd88.tar.gz* (if generated)